



MIKOŁAJ JAN PISKORSKI

Choosing Corporate and Global Scope

Introduction

When strategists analyze competitive strategies, they usually begin by looking at a single business unit operating in a single region and examine whether the business has a competitive advantage or disadvantage. Yet very few large and medium-sized corporations operate in a single industry or in a single geography. In fact, in 2005 the average US-based Fortune 500 company operated in four distinct industries, with some operating in more than ten. Moreover, firms frequently change the mix of industries in which they compete. Every year, the average large American corporation undertakes approximately three acquisitions, divests two businesses, and enters into six alliances.¹ Over a decade, such activities total over 100 transactions, amounting to significant changes of corporate scope. The same is true for global scope. An average US-based Fortune 500 firm derives 35% of its revenues from overseas operations. In fact, some companies, such as Boeing or Novo Nordisk, would never reach profitability if they did not compete globally.

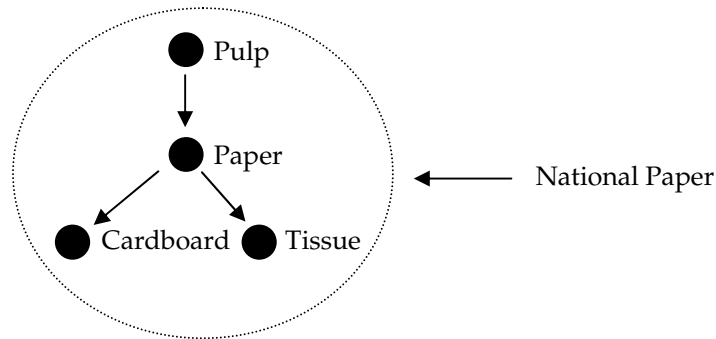
The objective of this note is to help you understand when firms should or should not enter into new industries or new geographies. As always, we will use the metric of competitive advantage and examine conditions under which presence in multiple industries or multiple geographies allows firms to increase the wedge between willingness to pay and costs. The logic we use to evaluate firms' presence in multiple industries is very similar to that which we use to evaluate presence in multiple countries. However, to be clear, we separate the two types of choices and start with the choice of industries.

Choosing Corporate Scope

To understand corporate scope, consider National Paper, a corporation comprising four business units: Pulp, Paper, Cardboard and Tissue. The corporation is schematically illustrated in Figure A, where four solid dots represent businesses owned by National Paper, and the dotted oval represents the boundary of the firm. The arrows inside the oval indicate flows of outputs between units.

Professor Mikołaj Jan Piskorski prepared the original version of this note, "Note On Corporate Strategy," HBS No. 705-449. This version was also prepared by Professor Mikołaj Jan Piskorski as the basis for class discussion. Parts of the global strategy discussion in this note were based on "Introduction to International Strategy" by David Collis and Jordan Siegel, HBS No. 706-481.

Copyright © 2007 President and Fellows of Harvard College. To order copies or request permission to reproduce materials, call 1-800-545-7685, write Harvard Business School Publishing, Boston, MA 02163, or go to <http://www.hbsp.harvard.edu>. No part of this publication may be reproduced, stored in a retrieval system, used in a spreadsheet, or transmitted in any form or by any means—electronic, mechanical, photocopying, recording, or otherwise—without the permission of Harvard Business School.

Figure A National Paper's multi-market presence

Source: Casewriter.

To characterize a corporation like National Paper, we use the concept of *corporate strategy*, which captures a set of choices that a corporation makes to create value through configuration and coordination of its multi-market activities. Thus, corporate strategy is different from competitive strategy. Corporate strategy pertains to the corporation as a whole, whereas competitive strategy pertains only to business units. Corporate strategy addresses choices about multi-market activities, while competitive strategy is a set of choices confined to a single market. However, there is an important point of integration between the two strategies: corporate strategy, if designed and executed well, will increase business units' competitive advantage over and above what they could otherwise achieve. When this occurs, the business units benefit from *corporate advantage*.

Many corporate strategy choices can generate corporate advantage, but two are particularly important (and readily apparent in Figure A). First, National Paper has chosen to operate simultaneously in Pulp, Paper, Cardboard and Tissue Production. The corporation has also chosen not to operate in other industries, such as forestry, publishing or food processing. We use the term *corporate scope* to denote the markets in which the corporation has chosen to maintain a competitive presence. Second, National Paper chose to maintain a presence in each of these markets by owning a business unit outright. There are a number of alternative ways in which the corporation could have maintained a presence. Instead of outright ownership, National Paper could have entered into a joint venture or alliance with another corporation in order to serve a particular market. Alternatively, it could have simply signed a long-term contract with a firm operating in that market.²

How would you evaluate National Paper's corporate strategy? Does it really increase the competitive advantage of its individual business units over and above what they could achieve otherwise? If not, what should change? Should National Paper exit the pulp industry to narrow its scope, or should it acquire control over forest holdings to control inputs for the pulp business? If scope expansion is warranted, should National Paper actually own a forest, or would it be better off to sign a long-term contract? To answer these two questions we employ two tests.

First, to assess the choice of markets in which a corporation maintains competitive presence, we will employ the better-off test:

Does the presence of the corporation in a given market improve the total competitive advantage of business units over and above what they could achieve on their own?

A negative answer to this question suggests that the corporation should exit the market. An affirmative answer means that a simultaneous presence in multiple industries is beneficial. It does not imply, however, that the corporation should necessarily own a business serving a given market.

To examine whether the corporation should own a given business, we will apply a second test — the ownership test:

Does ownership of the business unit produce a greater competitive advantage than an alternative arrangement would produce?

Only when both tests are satisfied should we conclude that a corporation made an appropriate decision to own a business unit in another industry. A positive answer to the better-off test accompanied by a negative answer to the ownership test tells us that the corporation should maintain a competitive presence in an industry but by some means other than full acquisition.³

The logic behind these tests is straightforward, but it is very difficult in practice to establish whether the conditions required by each test are satisfied. For starters, we are asked to compare a situation we can observe to one we cannot. For example, if we look at National Paper's Pulp unit, which is already owned by the corporation, we have to estimate what the costs and willingness to pay of this business unit would be if it were separated from the corporation. Since it is often hard to do this directly, we could identify a free-standing Pulp business unit and compare its costs and willingness to pay to the unit owned by National Paper. It is not always clear that two such businesses are perfectly comparable, however; they might possess different resources or have different product mixes, for instance.⁴ To overcome these problems, we will identify and discuss some circumstances that commonly cause the better-off and ownership tests to be satisfied.

Horizontal Diversification

We begin to examine these circumstances in the context of *horizontal diversification*. Horizontal diversification is simultaneous ownership of two or more units that utilize a similar set of tangible and intangible resources. In the case of National Paper, the Cardboard and Tissue units are horizontally related in that both utilize the outputs of the Paper unit.

The Better-Off Test

Economies of scope One of the key conditions that will satisfy the better-off test in a context of horizontal diversification is the presence of *economies of scope*. Economies of scope exist if the total cost of producing two goods jointly is less than the combined cost of producing each product separately. Formally, if C denotes the total cost of production, Y_1 is the output of product 1 and Y_2 is the output of product 2, then economies of scope will be present if

$$C(Y_1, Y_2) < C(Y_1, 0) + C(0, Y_2)$$

Economies of scope most frequently arise when a corporation owns a multi-purpose asset that is not fully utilized in production of a single product.⁵ Such assets can be tangible, such as manufacturing plants or distribution channels, or intangible, such as skills or research-and-development-generated knowledge that can support multiple products.

Consider an example from the automotive industry, where firms often need to invest in large and expensive stamping facilities, which exhibit economies of scale. Such facilities can stamp auto bodies as well as light-truck bodies. Suppose that neither autos nor light trucks generate enough market demand to exhaust the economies of scale. If the facilities produced only auto bodies or only light-truck bodies, they would never reach the volume required for maximum efficiency. When both body types are produced jointly, however, the facilities can reach sufficient volume. As a result,

production costs will be lower than they would be if the two body types were manufactured separately. Consequently, joint production of auto bodies and light truck bodies allows the two businesses to gain greater competitive advantage than auto manufacturers could achieve if they produced auto bodies and light trucks separately.

Similar analysis can be applied to intangible assets. To return to the National Paper example, if knowledge about efficient production of cardboard is applicable to the production of tissue, the Cardboard unit can transmit ideas about efficient methods to the tissue unit. As a consequence, the tissue unit will benefit from lower production costs, which will in turn improve its competitive advantage.

Cross-selling benefits When economies of scope exist, horizontal diversification generates cost advantages. Horizontal diversification can also produce willingness-to-pay advantages. Formally, if WTP denotes willingness to pay, then cross-selling benefits exist if

$$\text{WTP}(Y_1, Y_2) > \text{WTP}(Y_1, 0) + \text{WTP}(0, Y_2)$$

Cross-selling benefits are most likely to arise in one-stop-shop situations—particularly when customers want a single point of contact for sales, service and other support. For example, IBM's corporate strategy of providing a complete solution to firms' IT needs, including hardware, software and service, is based on this benefit. Cross-selling benefits also arise when reputation is important. Consider Honda's automobile, motorcycle and power-equipment businesses. To the extent that customers have similar needs and preferences in cars, motorcycles and lawnmowers—high reliability, high durability, high fuel efficiency, and low emission levels—the three businesses have an opportunity to exploit Honda's reputation for reliable, durable, and fuel-efficient machines.⁶

The Ownership Test

Cross-selling benefits explain joint sales, but not why these sales have to be undertaken by a single corporation. Two separate organizations can contract to cross-sell their products, as airlines do with code-share agreements. Similarly, economies of scope explain joint production, but not why joint production must be confined within a single corporation. If it is possible to write a contract to share the inputs that yield economies of scope, joint production does not require multi-product organization. To return to our automotive example, an automobile manufacturer could own the stamping facilities and a separate truck manufacturer could contract with the auto maker to stamp the truck bodies it needs. Conversely, the truck maker could own the stamping facilities and the auto manufacturer could contract for car bodies to be stamped there. Or a third firm could own the stamping facilities and contract its services to both manufacturers.

To tap economies of scope or cross-selling benefits, it is often less costly to use a contract between two separate companies than to have one corporation that owns operations in both industries. The separate companies are spared the direct costs of the corporate office that monitors the diversified corporation. The separate companies also save indirectly. For example, when businesses are owned separately, they make their capital allocation decisions independently. In contrast, when the corporate office makes these decisions, a business unit may find itself with too little or too much capital. This might lead to substantial power differences between business units, causing business-unit managers to engage in games to counteract the power structure. To the extent that these behaviors undermine trust and cooperation between businesses, and to the extent that managers spend their time playing internal politics rather than trying to win in the marketplace, the corporation will suffer and the corporate advantage will be eroded.

However, contractual arrangements may themselves be very costly to use for some types of exchanges. In these cases, it may be better for a firm to endure the costs of ownership. The costs of contracting are often particularly high when firms try to share or trade intangible assets, such as manufacturing or management skills. Consider once again the Cardboard business unit whose excellent manufacturing skills are also applicable to tissue production. The Cardboard unit could advertise that its expertise has reduced its annual manufacturing costs by \$50 million and that it is willing to impart this knowledge to tissue manufacturers for \$10 million. The proposed deal potentially creates value for both parties, but the tissue producers do not know whether the Cardboard unit's skills will really reduce their manufacturing costs. Its claims might be worthless, or might even harm the manufacturing process.⁷

To protect themselves, the tissue firms might insist that the Cardboard unit explain its manufacturing secret and how it is implemented. If the Cardboard unit were to disclose this information, however, the tissue producers would learn what to do and no longer need to pay for the acquisition of the skills; they could simply claim to have come up with the idea themselves and proceed to implement it. In anticipation of this problem, the Cardboard unit is unlikely to share its knowledge prior to signing a contract. This dilemma is known as *the fundamental paradox of information*—the prospective purchaser cannot know its value until the information is revealed, at which point he has in effect acquired it without cost.⁸

Thus, the tissue manufacturers will not want to buy such assets without understanding their value, but the Cardboard unit will not disclose such information for fear that the tissue manufacturers will not pay for it. Consequently, market exchanges of skills and other intangible assets are likely to fail. These market failures create an opportunity for a corporation to arrange an exchange that would not occur through markets. If the Cardboard unit and the Tissue unit are jointly owned by a single corporation, as in the case of National Paper, manufacturing or managerial skills can be more easily transferred between the two units. The Cardboard unit can disclose information with confidence that it will receive appropriate compensation. In these circumstances, the ownership test is satisfied: joint ownership is required to fulfill the promise of the better-off test.

Many intangible assets will satisfy the ownership test, but not all will. Consider, for example, a blueprint for a special device or a formula for a chemical compound. Since the buyer is able in both cases to recognize the value of the asset, it is possible for buyer and seller to write a contract for market exchange. In such cases, the ownership test will fail and integration into the buyer's corporation will not be necessary. In deciding whether the ownership test is satisfied, therefore, you should focus not on the nature of the good exchanged but on the presence or absence of obstacles to a contractual relationship.

Vertical Integration

We now turn to the better-off and ownership tests in a context of *vertical integration*. Vertical integration arises when one corporation owns business units that make inputs for other business units within the corporation. We typically refer to the producer of such inputs as *upstream* and the users as *downstream*. In the case of National Paper, the Pulp unit and the Paper unit are vertically integrated in that Paper utilizes the output of Pulp; Pulp is upstream and Paper is downstream. Similarly, the Cardboard and Tissue units are vertically integrated with the Paper unit, since both utilize paper pulp as inputs; in this case, Paper is upstream and the Cardboard and Tissue units are downstream.

The Better-off Test

Relationship-specific investments A key condition that tends to satisfy the better-off test in vertical integration is the presence of relationship-specific investments. Such investments arise when business units in a vertical relationship tailor their assets to exchanges with each other in order to reduce production costs or boost the willingness of customers to pay for their goods. Consider, for example, a cardboard producer that utilizes ordinary paper pulp, but subjects it to a costly process to produce high-margin cardboard. Suppose that this process could be made substantially cheaper if the cardboard producer invested in specialized equipment. However, the specialized equipment requires a supply of high-grade pulp not widely available on the market. To make its investment worthwhile, the cardboard producer needs to find at least one paper-pulp producer willing to invest in facilities to produce high-grade paper pulp. Once such a paper-pulp producer is identified and the specialized investments are made on both sides, the costs of production will fall without affecting cardboard customers' willingness to pay. This increase in added value will be distributed between the paper-pulp producer and the cardboard producer and thus contribute to each company's competitive advantage.

To distribute the added value, the cardboard producer may promise the paper-pulp producer to pay a high rate for pulp over time so that the paper producer can recoup its investment. However, such an agreement is problematic. Once the paper-pulp producer builds the new facilities, the cardboard producer can renege on its promise and refuse to pay more than the marginal cost for the specialized paper pulp. Since no other businesses want the specialized paper pulp, the paper-pulp producer will have no option but to sell its products at the marginal cost, failing to recoup its original investment. Foreseeing this problem, the paper-pulp producer may refuse to invest in production of specialized pulp; the paper-pulp producer's fear of being taken advantage of will preclude an investment that could enable both producers to improve their margins.

Joint ownership of paper and cardboard production might solve this problem. Joint ownership by a single corporation would assure the paper-pulp business unit that the cardboard unit will not take advantage of a relationship-specific investment. Consequently, it is more likely to make such an investment, enabling the cardboard business unit to obtain higher prices from its customers. Thus, joint ownership of paper and cardboard production will contribute to the competitive advantage of the vertically integrated corporation.

Downstream free-riding A related condition likely to satisfy the better-off test occurs when firms free-ride on the efforts of their competitors. Consider the relationship between a supplier and retailers of consumer goods. Suppose that there are two types of retailers: one provides customers thorough information on the product and excellent customer service; the other merely sells the product. The no-frills retailer has a clear cost advantage and can charge a lower price. Customers are thus likely to seek product information from a high-quality retailer but ultimately purchase from the no-frills retailer. As a consequence, the high-quality retailer will be disadvantaged, and will be likely to drop its services. This decision could hurt the manufacturer by decreasing overall demand for the product. Thus, in order to preserve customer service, the manufacturer might have to merge into retailing.⁹

The Ownership Test

The two conditions discussed above satisfy the better-off test and help us understand when a firm should have some presence in more than one industry along the vertical chain. However, as in the case of horizontal diversification, passing the better-off test is insufficient to conclude that a firm should actually own the upstream or downstream business. As before, there may exist significant

organizational costs to integrating two units into a corporation that may override some of the benefits. For example, when businesses are not integrated, they must compete with other firms in their markets and thus continually strive to provide the best-quality products at the lowest prices. When businesses are integrated, in contrast, the corporate office can mandate transfers of resources between business units at pre-arranged transfer prices. A consequence may be weakened incentives for the upstream business units to be the most efficient producers. If this is the case, the firm is advised to use a contract rather than integration. Only when contracts are costly to write, monitor, and enforce would we expect a firm to acquire along the vertical chain. In fact, as we will see below, very few conditions that pass the better-off test also pass the ownership test.

Relationship-specific investments Recall our discussion of relationship-specific investments between a paper-pulp producer and a cardboard producer. Such investments could improve the competitive advantage of both firms but create vulnerability to opportunistic renegotiation, making the investments unlikely. We may consequently conclude that the firms should integrate to preclude renegotiation and to ensure that the investments are undertaken. A simpler solution, however, may be a long-term contract between the parties. If the firms can specify the price at which they will exchange and create safeguards against future opportunistic renegotiation, they can assure each other that there will be no such renegotiation. Many such contracts exist in practice. For example, transactions between coal producers and coal-burning plants, characterized by substantial relationship-specific investments, are typically governed by long-term contracts.¹⁰ Both parties can enforce detailed contracts that specify prices, quality, and terms of delivery. If specifying these basic parameters in a contract is costly or even impossible, the ownership test is satisfied. In such cases, joint ownership of upstream and downstream may be the only way to encourage relationship-specific investments.

The broader institutional environment will have a major impact on whether this test will be satisfied. Firms that operate in a country where contracts are easy to enforce through well-developed institutions, such as courts, are likely to use contracts as frequently as they can. In many developing countries, however, these institutions may be weak, corrupt, or absent. In such environments, each company may fear that the other will fail to honor its commitments. They may thus be wary of entering into contracts, even though they would not hesitate to do so in a different legal environment. In these contexts the ownership test will be satisfied. One important implication is that firms in developing countries may tend to have a broader scope than their counterparts in developed countries.¹¹ Thus, National Paper of India, China or Poland may be broader in scope than its American equivalent. By integrating exchanges into the firm, the broadly diversified corporations typical of those countries protect their business units from the perils of market exchanges in uncertain legal environments, and thus secure a corporate advantage for their business units.

Downstream free-riding A similar analysis can be applied to downstream free-riding. Vertical integration seems to be an effective means of preventing the free-riding problem, but a host of contractual solutions could be employed instead. The upstream producer could force all retailers to charge the same price. (This practice is generally illegal in the United States.) Another alternative is to do business only with retailers that provide the required services. A less extreme version of this strategy is to provide better treatment—faster restocking or joint advertising campaigns—to retailers that provide service. If none of these options is feasible, the ownership test is satisfied and the firms should integrate.

Choosing Global Scope

Thus far, we have explored the two tests that firms should undertake before they acquire a business unit in a different industry. We will now examine the tests that firms should employ when deciding to expand their global scope by entering another country. Few big firms can afford to ignore the global aspect of competition these days, but the decisions to expand globally are by no means straightforward. Consider, for example, National Paper's decision to become International Paper. Should the company establish a wholly-owned subsidiary in another country, say Sweden, and replicate its US-based business model? Or should National Paper only acquire forest holdings in Brazil which will provide inputs to the US business? Or maybe the company would be even better off only establishing a new sales organization in China with production still in the US? If so, should production and R&D be shifted there too? Finally, maybe the right way to think about global scope is not in a country-by-country fashion, but in terms of a portfolio of countries. To answer these questions we use exactly the same logic we used to appraise firm diversification decisions, employing the same better-off and ownership tests. As you will see, the analogy between the two types of decisions is so close that we even word them in the same way.

Thus, to appraise whether a firm should enter into another country, we use the better-off test, which asks:

Does the presence of the corporation in a given geographic market improve the total competitive advantage of business units over and above what they could achieve on their own?

A negative answer to this question suggests that the corporation should not operate in a different country. An affirmative answer means that a simultaneous presence in multiple geographies is beneficial. It does not imply, however, that the corporation should necessarily own assets in that country. To examine whether the firm should own assets in another country, we apply the ownership test, which asks:

Does ownership of the business unit in a geographic market produce a greater competitive advantage than an alternative arrangement would produce?

We should only conclude that a firm made an appropriate decision to enter another country when both tests are satisfied. A positive answer to the better-off test accompanied by a negative answer to the ownership test suggests that import or export of relevant resources or other contractual arrangements may achieve the required outcome. Negative answers to both tests suggest that the firm should not expand its global scope. We now turn to examine key conditions under which we expect the better-off and the ownership tests to be satisfied.

The Better-off Test

Factor cost differences One of the first global strategies ever invented entails locating overseas to exploit lower prices of inputs in other countries. Historically, the resource that firms sought to exploit was natural resources. The oil giants grew by acquiring drilling rights and reserves in countries with oil fields. The big agricultural firms, like the United Fruit Company, developed huge overseas plantations to exploit the comparative advantage of tropical countries in growing fruit. These days firms tend to rely on overseas locations to employ lower-cost labor than is available in their home market. As competition in output markets has matured, firms have grown increasingly sophisticated in their use of factor cost differences and have started to locate their activities in

numerous countries. This allows them to choose among countries with lower wage rates, avoid dock strikes in one country that could halt production, make sure they do not miss out on a breakthrough technology developed in another country, and so on. Once firms establish their presence in multiple countries, they can change the source of their inputs almost in real time in order to obtain the lowest input costs. Becton Dickinson, for example, the manufacturer of blood containers, shifts the production of its standard global products between plants in three different regions in order to concentrate on the current low-cost location.¹² Procter and Gamble does the same with its detergent formulations.

Economies of scale In addition to sourcing outputs overseas, firms can also achieve a competitive advantage by selling their goods abroad and thereby achieving a much larger scale. While some US firms, like Caterpillar and Boeing, currently thrive using this approach, it was Japanese corporations that first exploited this strategy thoroughly. In a series of industries, from motorbikes to televisions to machine tools, Japanese firms in the 1960s and 1970s developed what became recognized as true global strategies.¹³ In the extreme, this strategy exploits global scale economies by developing, manufacturing, and selling a single product everywhere around the world.¹⁴ While the product might not be optimal for every country, the cost advantage from global production volumes allowed firms like Honda or Matsushita to achieve market leadership in many countries even with a limited array of standard products. It is worthwhile to note that this strategy does not require a firm to have a competitive advantage at home before it goes abroad. It is the international activity that allows the firm to generate a competitive advantage.

Exploiting knowledge around the world The third way in which global firms can establish a competitive advantage over local firms is by selling the original product in a new country, or replicating the domestic strategy in the new market. For example, most multinationals in the consumer goods industry emerged this way. They began by exporting their original product overseas; they then set up a distributor; finally, they established a full subsidiary and began to manufacture as well as sell in that country.¹⁵ The value created by such a strategy lies in replicating a strategy that has been demonstrated to work well domestically, in a country that was previously missing that particular product or service. The firm arbitrages a product or a strategic capability into a new market. That same firm would also be in a position to transfer a process innovation, perhaps developed in one plant in response to specific material shortage in that country, to the rest of its plants. Similarly, if the firm has many R&D centers, it can be an insider with access to product innovations or even fashion trends from those many countries.¹⁶

The Ownership Test

Just as corporate strategy must explain why the firm has to be active in the new market rather than simply licensing or selling its valuable resource to a firm that already competes in that market, so a firm that competes internationally also has to address the question of why it cannot simply license or sell its valuable resource to an indigenous firm (or buy it from an indigenous firm in the case of supply side scope economies). In other words, why does the firm have to become a multinational? Unilever, for example, might have developed a wonderful new detergent technology and a great advertising campaign to promote the product in the UK, but why can it not just license that product and marketing program to a US firm? Why should Unilever itself have to compete in the US market to realize the rents that the new product generates? By doing so, the firm would save tremendous overhead costs associated with managing inside the firm across geographic borders.

As with corporate strategy, the answer to this question lies in identifying the market failure that prevents efficient contracting between the parties in different countries. More generally, however, the

same factors that underlie market failures across product boundaries are also relevant across geographic boundaries.¹⁷ Whenever the asset being transferred across countries is intangible or difficult to define, or if the asset is information whose value is transferred when it is revealed, contracts can be very costly or even impossible to write. This explains why Microsoft is loath to license its code to a Chinese firm to convert to Mandarin and sell in China.¹⁸ Similarly, when an asset is specific to a particular relationship the risk of hold-up by either party leads to market failure. This explains why Alcan owns bauxite mines in Jamaica to supply its alumina refineries. Since the refinery's equipment has to be matched to the specific grade of ore from a particular mine, the two stages of production have to be owned together. These concerns are accentuated by the fact that international contracts are difficult to write because so many of the preconditions for the effective governance of market transactions are missing in many countries.¹⁹ The rule of law, a mutual trust in business relationships, and effective contract enforcement mechanisms are often missing or incomplete, in many countries. If a firm is to compete in such countries, it will have to do so itself, or face enormous risks, such as the expropriation of its assets or the lack of effective recourse against partners who renege on contractual terms. As a consequence, firms often choose the ownership route.

Conclusion

By now you should understand the basic principles underlying value-creating corporate and global diversification. A corporation contemplating a competitive presence in another industry or another country should examine the decision against the better-off and ownership tests. In a context of horizontal diversification, economies of scope and cross-selling benefits are most likely to lead to a conclusion that the better-off test is satisfied. In a context of vertical integration, the better-off test is most likely to be satisfied in the presence of relationship-specific investments or the possibility of downstream free-riding. To pass the ownership test for both types of diversification, the firm should examine if there are significant obstacles to negotiating and enforcing a contract. If there are no such obstacles, a contract is preferable as it avoids ownership costs. However, when certain intangible assets are involved, full acquisition is likely to yield greater benefits. Similarly, to justify its presence in numerous countries, a firm must also pass the better-off test. This is most likely if the firm can benefit from sourcing cheap inputs, from economies of scale, or through introduction of goods or resources from its home country to the new market. Since the costs of global organization are likely to be large, firms should use contractual arrangements unless contracts are costly to write, monitor, and enforce. In those circumstances, the ownership test is fulfilled, and the firm should own its overseas operations.

Endnotes

¹ All calculations in this paragraph are based on notewriters' research and cover the period of 1995 to 2005.

² For full analysis of the various decisions that lead to corporate advantage, see Collis and Montgomery (1998).

³ Some scholars have argued that in choosing corporate or geographic scope we should also employ the 'cost of entry' test. Such test requires that the anticipated benefit of expanding corporate scope is larger than the cost of acquiring resources needed to produce it. As this test applies to any strategically sound decision, we omit it here for brevity of exposition.

⁴ For example, Villalonga (2004) provides evidence that stand-alone business units may be of higher quality, suggesting that straightforward comparisons of stand-alone businesses and corporate business units may be misleading in assessing the appropriate scope of a firm.

⁵ See Panzar and Willig (1981) and Bailey and Friedlaender (1982) for a complete discussion of factors that yield economies of scope.

⁶ Montgomery and Wernerfelt (1992) provide a detailed analysis of the effects of reputation.

⁷ The contractual problem described here is the adverse-selection problem. Akerlof (1970) first recognized and illustrated this problem in the market for used cars. Adverse selection stems from information asymmetry: to employ Akerlof's example, the owner of a used car has access to far more information about its quality than does any potential buyer. Without accurate information on the quality of a given car, buyers will tend to pay the same amount for all cars with the same observable attributes. Only owners whose estimation of their cars' value is less than the average price will sell their cars, while owners of high-quality cars will not find it attractive to enter the market. Consequently, only poor-quality goods will be available for sale, and the market for high-quality goods will fail.

⁸ See Arrow (1971), page 152.

⁹ The third condition likely to satisfy the better-off test arises when two firms in a vertical relationship each possess market power. Because each firm faces a downward-sloping firm-specific demand, both will price at a markup over marginal cost. This markup over marginal cost raises prices and reduces the equilibrium output. Since both firms are pricing above marginal cost, the price increase and output reduction are incurred twice. If the firms merged, however, the supplier would cease to capture surplus and instead transfer goods and services at marginal cost. This scenario would allow for lower output prices and expansion of output, both of which increase the competitive advantage of the buyer. The final condition arises through vertical foreclosure. Consider a market whose supply of inputs is competitive before a merger. Suppose that, after a merger, the upstream division of the now-integrated firm refuses to supply inputs to rivals of its downstream division. This foreclosure of rivals means that other suppliers will face less competition. As a result, they may be able to increase profits by raising their input prices to the unintegrated downstream firms. These higher prices benefit the downstream division of the vertically integrated firm. If rivals' input costs increase, they will be forced to reduce production and to raise the prices they charge in the downstream market. This reduction in competition allows the downstream division of the integrated firm to increase its market share and its price. Thus, the profits of the vertically integrated firm can rise, even if no production-efficiency benefits flow from vertical integration. As this description illustrates, the rationale behind vertical foreclosure is complex. Thus, although vertical foreclosure has elicited substantial academic interest, there is little real-life evidence that vertical foreclosure is an important driver of vertical integration (Ordover, Saloner, and Salop 1990).

¹⁰ See Joskow (1987) for details.

¹¹ See Khanna and Palepu (2000).

¹² George S. Yip, *Total Global Strategy* (Prentice Hall Trade, 1992).

¹³ Accordingly, this strategy is commonly referred to as the global strategy, Michael Porter, "Global Strategy: Winning in the World-Wide Marketplace," in *The Portable MBA in Strategy*. Liam Fahey and Robert M. Randall (eds.), New York: John Wiley & Sons, Inc., 1994.

¹⁴ Ted Levitt was the most famous early proponent of this approach in his classic work "The Globalization of Markets," *Harvard Business Review* (May 1983).

¹⁵ Ray Vernon, "International Investment and International Trade in the Product Cycle," *Quarterly Journal of Economics* 80, (1966) pp. 190-207.

¹⁶ Christopher A. Bartlett and Sumantra Ghoshal, *Managing Across Borders*, (Harvard Business School Press, 1989).

¹⁷ Bharat Anand, Tarun Khanna and Jan Rivkin, "Market Failures," HBS No. 700-127 (Boston: Harvard Business School Publishing, 2000).

¹⁸ Tarun Khanna, "Microsoft in the People's Republic of China—1993," HBS No. 795-115 (Boston: Harvard Business School Publishing, 1995).

¹⁹ Tarun Khanna and Krishna Palepu, "Spotting Institutional Voids in Emerging Markets," HBS No. 106-014 (Boston: Harvard Business School Publishing, 2005).¹ Bharat Anand, Tarun Khanna and Jan Rivkin, "Market Failures," HBS No. 700-127 (Boston: Harvard Business School Publishing, 2000).

References

- Akerlof, George. 1970. The Market for 'Lemons': Qualitative Uncertainty and the Market Mechanism. *Quarterly Journal of Economics* 84:488-500.
- Arrow, Kenneth. 1971. *Essays in the theory of risk bearing*. Chicago: Markham.
- Bailey, Elizabeth, and Ann Friedlaender. 1982. Market Structure and Multiproduct Industries. *Journal of Economic Literature* 20:1024-1049.
- Collis, David J., and Cynthia A. Montgomery. 1998. *Corporate Strategy: A Resource-Based Approach*. Boston: Irwin/McGraw-Hill.
- Joskow, P. 1987. "Contract Duration and Relationship-Specific Investment: Empirical Evidence from Coal Markets." *American Economic Review* 77:168-85.
- Khanna, Tarun, and Krishna Palepu. 2000. Is Group Affiliation Profitable in Emerging Markets? An Analysis of Diversified Indian Business Groups. *Journal of Finance* 55:867-891.
- Montgomery, Cynthia A., and Birger Wernerfelt. 1992. Risk Reduction and Umbrella Branding. *Journal of Business* 65:31-50.
- Ordover, Janusz A., Garth Saloner, and Steven C. Salop. 1990. Equilibrium Vertical Foreclosure. *American Economic Review* 80:127-142.
- Panzar, John C., and Robert D. Willig. 1981. Economies of scope. *American Economic Review* 71:268-272.
- Villalonga, Belen. 2004. Does Diversification Cause the 'Diversification Discount'? *Financial Management* 33:5-27.